

A User-Configurable Navigation System for Wideness and Turn-Aware Routing

Kousik Kumar Dutta¹ Venkata M. V. Gunturi²

¹Department of Computer Science and Engineering, Indian Institute of Technology Ropar, India

²Department of Computer Science, Loughborough University, UK

FlexiRoute | IEEE MDM 2026 | System Demonstration

Motivation

Long / towed vehicles: greater turning radius and off-tracking.

Beyond fastest / shortest: narrow roads and right-across-traffic turns can be difficult.

Time changes width: clearway / parking restrictions temporarily free road space.

Wider roads and fewer difficult turns, within a travel-time budget?

Your query, three preferences

Input: source s , destination d ; departure window $[t_S, t_E]$; time budget B ; preference.

WIDE

Maximize road wideness percentage.

TURN-SAFE

Minimize topology-based right-turn count.

BALANCED

Explore wideness / right-turn Pareto trade-offs.

All routes respect the budget. Wideness = wide distance / total distance. Right-turn count excludes single-exit junctions and roundabouts.

Why FlexiRoute is different

Objective-decoupled exploration. Reuse candidates across all three modes.

Temporal width + turn semantics. Time-dependent width meets topology-based turns.

Explainable trade-offs. Pareto routes, highlighted turns and departure-time effects.

London demonstration setup

288,016

nodes

744,610

edges

OpenStreetMap-synchronized visualization; temporal width and topology-based turns.

Synthetic clearway schedules emulate width changes, not measured ground truth.

Interactive latency

38 ms

22 ms

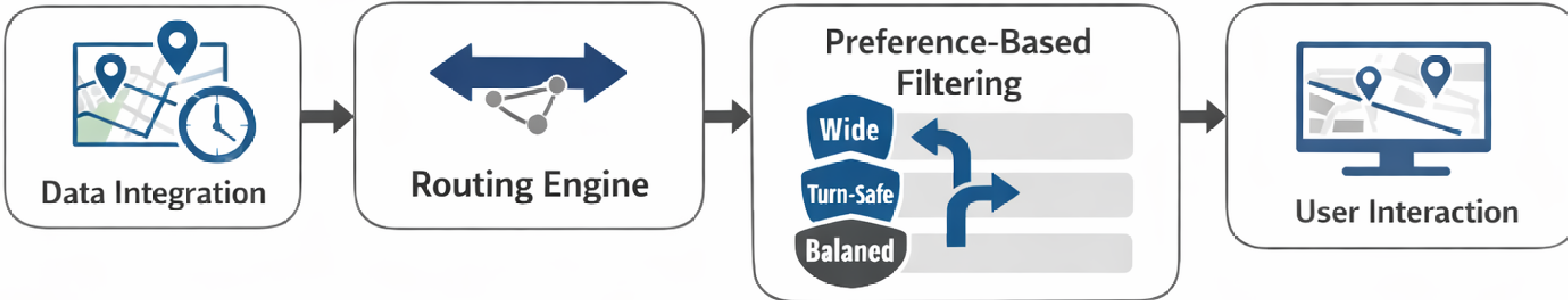
Query	Labels	Initial	Switch
Q1	2,800	183 ms	38 ms
Q2	1,400	168 ms	22 ms

Switch modes by re-ranking cached labels, without graph search.

Demo queries: Q1 (budget 50, 15 nodes); Q2 (budget 60, 50 nodes).

Intel Xeon W-2155 3.30 GHz; 10 cores / 20 threads; 32 GB RAM.

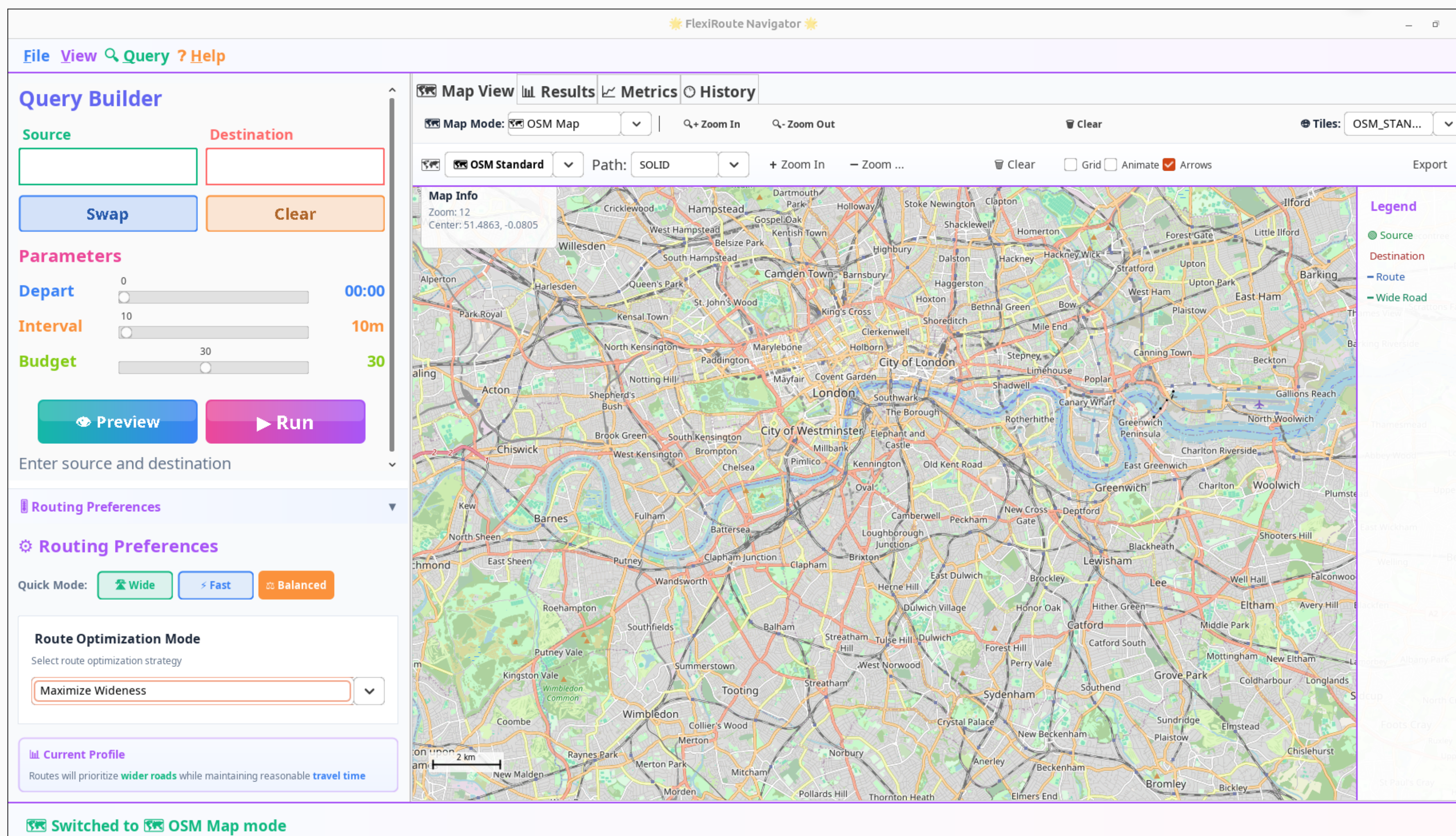
FlexiRoute: search once, explore many preferences



1. Data integration
2. Adaptive bidirectional A* label generation (interval-aware)
3. Preference filtering / re-ranking
4. User interaction

Search once → reuse feasible labels → switch preferences

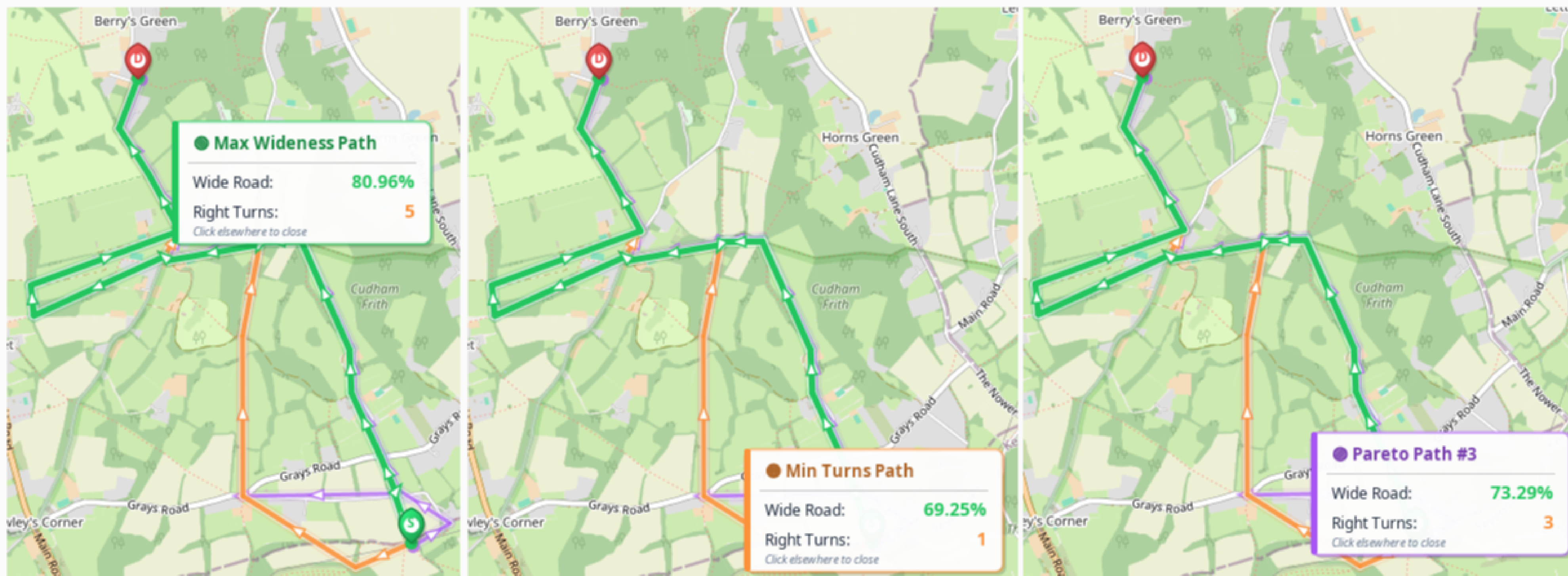
Interactive demo



Interactive query builder and OpenStreetMap-based visualization.

Select source / destination → set departure window and budget
Choose mode → run once → instantly explore alternatives

One caravan journey, several Pareto alternatives



WIDE
~80% wideness
5 right turns

TURN-SAFE
~69% wideness
1 right turn

OTHER PARETO
~73% wideness
3 right turns

Caravan scenario: depart within 8 AM–2 PM. Recommended: 9:19 AM, exploiting temporal road widening.

No single route is universally best: compare wider roads with fewer difficult turns.

Pareto-optimality is relative to the generated feasible label set.

SEARCH ONCE

Reuse feasible labels across preferences.

TIME MATTERS

Departure time can change usable width.

EXPLICIT TRADE-OFFS

Wideness vs. turns, within a time budget.



Code / Artifact

github.com/kousik-kr/FlexiRoute
youtu.be/ID4RadWAGns



Demo Video